



Contents

1 Overview	3
2 Uniform Resource Identifiers	3
3 Data Collection and Usage Interfaces.....	4
3.1 Get patient data	4
3.2 Get patient logged events.....	15
3.3 Subscription/Notification interfaces.....	19
3.3.1 Patient notifications (AVX to IA).....	20
3.3.2 Integrator responses with corresponding AVX behavior.....	21
4 Appendix A. Supported system events.....	23
4.1 Astral system events	23
4.2 Stellar system events	24
5 Appendix B. Supported alarm types	26
5.1 Astral alarms	26
5.2 Stellar alarms.....	29
5.3 AirCurve 10 ST-A alarms	30
6 Appendix C. Supported devices	31
6.1 S9 devices	31
6.2 All other devices	31

1 Overview

This AirView Exchange (AVX) technical specification only covers Data Collection and Usage APIs exposed by AVX, which are intended for use by integration partners.

Before you can use the APIs outlined in this document, you must first acquire the patient's EasyCare Number (ECN). Refer to the *AVX API Technical Specification for Patient Management* for instructions on how to get a patient's ECN.

The information in this technical specification supplements the *AVX API Technical Specification for Patient Management*. It aligns with Release 4.38 of AirView for the Americas and the Asia-Pacific region.

2 Uniform Resource Identifiers

The following table depicts the hierarchical structured design of AVX information. It also summarizes the HTTP calls that are available at each level.

URIs are specified in a manner that is consistent with RFC 6570 URI Template, and are case sensitive.

URI	Method	Description
https://<host>/v1/patients/{ecn}/data/sessions/{date}	GET	Returns patient data for the specified day. Results include all device data for the specified day, regardless of when this data is received by AirView. Note: Results return only one day of data per call. For example, this can be called to back-load data for specific days. See "3.1 Get patient data" on page 4.
https://<host>/v1/patients/{ecn}/data/sessions?receivedOn={date}	GET	Returns all patient data that AirView received on a specified day. Note: Results can contain more than one day's worth of data. For example, if a data card was uploaded with 30 days of new data on March 30, this call would return all 30 days' worth of data as the result (March 1–30). See "3.1 Get patient data" on page 4.
https://<host>/v1/patients/{ecn}/data/sessions?noOfDays={days}&endDate={date}	GET	Returns all patient data for a specified date range. See "3.1 Get patient data" on page 4.
https://<host>/v1/patients/<ecn>/data/loggedEvents	GET	Returns the last five days of alarm, system and settings change events. See "3.2 Get patient logged events" on page 15.
https://<host>/v1/subscriptions/patients/{ecn}/data/sessions	POST	Used to subscribe to patient data notifications. See "Subscribing to a patient record for new summary data" on page 19.

3 Data Collection and Usage Interfaces

The following interface methods allow patient data to be retrieved.

3.1 Get patient data

The API supports three calls that are used to retrieve patient data, as detailed in the following table. To better understand the differences between various `getData` APIs, refer to the *AVX Data Collection and Usages Best Practices Guide*.

Note: Patients with an active and inactive status will be returned. If you would like to filter by status, refer to the *Retrieve a patient's details* section in the *AVX API Technical Specification for Patient Management*.

Property	Value
Method	GET
URLs	<code>https://<host>/v1/patients/{ecn}/data/sessions/{date}</code> AVX returns the data session for the patient based on the specified <i>ecn</i> and <i>date</i> . Note: Date format in YYYY-MM-DD. Typically used for retrieving session data for a single specific day (eg, retrieve device data for a single day for historical purposes). <code>https://<host>/v1/patients/{ecn}/data/sessions?noOfDays={days}&endDate={date}</code> AVX returns data sessions for patient <i>ecn</i> for <i>noOfDays</i> ending on <i>endDate</i> inclusive. Note: <i>noOfDays</i> parameter may be between 0 and 90 and date format in YYYY-MM-DD. Typically used for retrieving a range of historical device data (eg, retrieve device data for a historical period). <code>https://<host>/v1/patients/{ecn}/data/sessions?receivedOn={date}</code> AVX returns data sessions for patient <i>ecn</i> that arrived on the provided <i>receivedOn</i> date. Note: Date format for <i>receivedOn</i> is YYYY-MM-DD. Best used when called daily for synchronization purposes. This returns all data that was added to AirView on a specific day for patient <i>ecn</i> . This can return more than a single day of data (eg, a card upload where 30 days of data is uploaded to AirView on the <i>receivedOn</i> date).

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" Object – See table on the following page

The following fields are included in the JSON “Response” Object if they are reported and relevant to the device and mode used in the session. **Note:** If a device does not support a metric, that metric is not included in the response.

Field Name (JSON Name)	Type/Format	Description	From Rev:
responseCode	string	1004000	2.2
responseDesc	string	Unspecified – Debugging purposes only	2.2
patientData ()	object	Contains patient data	2.2
ecn	string	Patient unique identifier	2.2
totalUsageHoursOnDevice	string	Total usage duration in hours since last device reset	4.13
sessionData ([])	object array	Session data for the patient	2.2
receiptTime	string	System time of data receipt to AirView	2.2
sessionDate	string	The day this data is for	2.2
deviceID ()	object	Device used by this patient on this day	2.2
mode	string	Therapy mode that device was in on this day	2.2
device ()	object	Device ID information	2.2
serialNo	string	Serial number of the device used by the patient on this day	2.2
deviceType	string	Type of device used by the patient on this day	2.2
deviceTypeDesc	string	Description of the type of device used by patient on this day	2.2
usage ()	object	Usage statistics	2.2
duration	string	Total usage duration in minutes	2.2
maskOn []	string array	An array of mask on times in format: YYYY-MM-DD_HH:MM:SS	2.2
maskOff []	string array	An array of mask off times in format: YYYY-MM-DD_HH:MM:SS	2.2
programNumber	string	The program this data applies to. Note: Only applies to Astral™ devices.	4.1
programUsage	string	Total usage of this program in minutes. Note: Only applies to Astral devices.	4.1
set ()	object	Settings. Note: Setting only included if applicable to the mode.	2.2
press	string	The set pressure in cmH ₂ O	2.2
startPress	string	The start pressure in cmH ₂ O. Note: Only available to Stellar™ devices.	4.2
maxPress	string	The maximum allowable pressure in cmH ₂ O	2.2
minPress	string	The minimum allowable pressure in cmH ₂ O	2.2
EPRTType	string	Expiratory Pressure Relief type	2.2
EPRLLevel	string	Expiratory Pressure Relief level (relevant if EPRTType not OFF)	2.2
ODThreshold	string	Desaturation Rule Threshold setting	3.6
iBREnable	string	Intelligent Backup Rate Enable setting	3.6
tgtPatientRate	string	Target Respiratory Rate/Target Patient Rate setting in breaths per minute	3.6
EPAPAutoEnable	string	Auto EPAP Enable setting	3.6

Field Name (JSON Name)	Type/Format	Description	From Rev:
IPAP	string	Inspiratory Positive Airway Pressure setting in cmH ₂ O	3.0
maxIPAP	string	Maximum IPAP setting in cmH ₂ O	4.4
EPAP	string	Expiratory Positive Airway Pressure setting in cmH ₂ O	3.0
startEPAP	string	Start EPAP pressure setting in cmH ₂ O. Note: Only available to Stellar devices.	4.2
autosetResponse	string	AutoSet Response Setting On (Soft)/Off (Standard)	3.0
minEPAP	string	Minimum EPAP setting in cmH ₂ O	3.2
maxEPAP	string	Maximum EPAP setting in cmH ₂ O	3.2
minPS	string	Minimum Pressure Support setting in cmH ₂ O	3.2
maxPS	string	Maximum Pressure Support/P Control Max setting in cmH ₂ O	3.2
easyBreathe	string	The Easy Breathe setting. May be one of: - On - Off	4.6
backupRate	string	Backup respiratory rate	28.0
circuit	string	Breathing Circuit Type setting. May be one of: - SingleValved - DualValved - SingleVented - MouthpieceCircuit	4.1
interface	string	Patient Interface Type setting. May be one of: - Invasive - Mask - Mouthpiece	4.1
mask	string	Mask Type setting on the device. Note: Only available to Stellar and Astral devices. May be one of: - FullFace - Nasal - NasalUltra - Pillows - Paediatric - InLineVent - Infant	4.1
pressureAssist	string	Pressure Assist setting in cmH ₂ O. Note: Also known as Pressure Control/P Control.	4.1
safetyTidalVolume	string	Safety Tidal Volume setting in mL	4.1
tidalVolume	string	Tidal Volume setting in mL	4.1
Ti	string	Ti setting in seconds. Note: Only available to Stellar and Astral devices.	4.2

Field Name (JSON Name)	Type/Format	Description	From Rev:
tiMin	string	The Ti Min setting in seconds. Note: Only available to Stellar and Astral devices.	4.2
tiMax	string	The Ti Max setting in seconds. Note: Only available to Stellar and Astral devices.	4.2
riseTime	string	Rise Time setting in msec. Note: Only available to Stellar and Astral devices.	4.2
triggerSensitivity	string	Trigger Sensitivity setting. Note: Only available to Stellar and Astral devices. May be one of: • VHi • Hi • Med • Lo • VLo • Off	4.2
cycleSensitivity	string	Cycle Sensitivity setting. Note: Only available to Stellar devices. May be one of: • VLo • Lo • Med • Hi • VHi	4.2
respRate	string	Respiratory Rate/Backup Rate in breaths per minute. Note: Only available to Stellar and Astral devices.	4.2
tgtAlveolarVent	string	Target Alveolar Ventilation setting in liters per minute	4.2
triggerThreshold	string	Trigger Threshold setting in liters per minute	4.2
triggerType	string	Trigger Type setting. May be one of: • None (Off) • Absolute (Flow) • Relative (Pressure)	4.2
triggerTypeMouthPiece	string	Trigger Type Mouthpiece setting. May be one of: • Touch • High • Medium • Low • Off	4.7
cycleThreshold	string	Cycle Threshold setting in %	4.2
patientType	string	Patient Type setting. May be one of: • Adult • Paediatric	4.2
heightCm	string	Height setting in centimeters	4.2

Field Name (JSON Name)	Type/Format	Description	From Rev:
pressSupport	string	Pressure Support setting in cmH ₂ O. Note: Only available to Astral devices.	4.2
volumeBreathOptions	string	Volume Breath Options setting. May be one of: - TiOption (Ti) - PifOption (PIF)	4.2
flowShape	string	Flow Shape setting. May be one of: - Constant - Dec75% - Dec50% - Dec25%	4.2
manualBreath	string	Manual Breath enable setting	4.2
manualMagnitude	string	Magnitude setting in % when Manual Breath is enabled	4.2
peakFlow	string	Peak flow setting in liters per minute when PIF is selected as the volume breath option	4.2
flow	string	The fixed rate of air flow in liters per minute delivered to the patient. Note: Only available for High Flow Therapy (HFT) devices.	28.0
sigh()	object	Sigh settings	4.2
enable	string	The sigh enable setting	4.2
alert	string	The sigh alert setting when sigh is enabled	4.2
interval	string	The sigh interval setting in minutes when sigh is enabled	4.2
magnitude	string	The sigh magnitude setting in % when sigh is enabled	4.2
apneaVent()	object	The Apnea ventilation settings. Note: Only available to Astral devices.	4.2
respRate	string	The Respiratory Rate setting in breaths per minute for apnea ventilation	4.2
Ti	string	The Ti setting in seconds for apnea ventilation	4.2
peakInspFlow	string	The Peak Inspiratory flow (PIF) setting in litres per minute for apnea ventilation	4.2
pressureControl	string	The Pressure Control/P Control setting in cmH ₂ O for apnea ventilation	4.2
tidalVolume	string	The Tidal Volume setting in mL for apnea ventilation	4.2
alarmSettings()	object	The Alarm settings. Note: Only available to Astral devices.	4.2
apnea()	object	The Apnea Alarm settings	4.2
breathType	string	The Breath Type/Detection setting for apnea alarms. May be one of: - NoBreath - NoSpontBreath	4.2

Field Name (JSON Name)	Type/Format	Description	From Rev:
Interval	string	The Interval/T Apnea setting in seconds for apnea alarms.	4.2
Response	string	The Response setting for apnea alarms. May be one of: - AlarmOnly - AcvAlarm - PacvAlarm - Off	4.2
clinicalMetrics ()	object	Clinical metrics over the session	2.2
tgtIPAP ()	object	tgtIPAP metrics (Target inhalation positive airway pressure). Note: From v3.0, this reports IPAP metrics on S9™ bilevel devices also.	2.2
5	string	5% of target IPAP in cmH ₂ O	4.1
95	string	95% of target IPAP in cmH ₂ O	2.2
50	string	Median target IPAP in cmH ₂ O	2.2
max	string	Maximum target IPAP in cmH ₂ O	2.2
peakInspiratoryPressure ()	object	Peak Inspiratory Pressure	4.1
5	string	5 th Percentile in cmH ₂ O	4.1
95	string	95 th Percentile in cmH ₂ O	4.1
50	string	Median in cmH ₂ O	4.1
tgtEPAP ()	object	tgtEPAP metrics (Target exhalation positive airway pressure)	3.0
5	string	5% of target EPAP in cmH ₂ O	4.1
95	string	95% of target EPAP in cmH ₂ O	3.0
50	string	Median target EPAP in cmH ₂ O	3.0
max	string	Maximum target EPAP in cmH ₂ O	3.0
endExpiratoryPressure ()	object	End Expiratory Pressure	4.1
5	string	5 th Percentile in cmH ₂ O	4.1
95	string	95 th Percentile in cmH ₂ O	4.1
50	string	Median in cmH ₂ O	4.1
leak ()	object	Leak metrics	2.2
5	string	5% of leak in liters per second	4.1
95	string	95% of leak in liters per second	2.2
50	string	Median leak in liters per second	2.2
max	string	Maximum leak in liters per second	2.2
leakValved()	object	Valved Leak metrics	4.1
5	string	5% of valved leak in %	4.1
95	string	95% of valved leak in %	4.1
50	string	Median valved leak in %	4.1
respRate ()	object	Respiratory rate metrics	2.2
5	string	5% of respiratory rate in breaths per minute	4.1
95	string	95% of respiratory rate in breaths per minute	2.2

Field Name (JSON Name)	Type/Format	Description	From Rev:
50	string	Median respiratory rate in breaths per minute	2.2
max	string	Maximum respiratory rate in breaths per minute	2.2
ieRatio ()	object	ieRatio metrics (inhalation:exhalation ratio)	3.0
5	string	5% of I:E ratio	4.1
95	string	95% of I:E ratio	3.0
50	string	Median I:E ratio	3.0
max	string	Maximum I:E ratio	3.0
minuteVent ()	object	Minute ventilation metrics	3.0
5	string	5% of minute ventilation in liters per minute	4.1
95	string	95% of minute ventilation in liters per minute	3.0
50	string	Median minute ventilation in liters per minute	3.0
max	string	Maximum minute ventilation in liters per minute	3.0
tidalVol ()	object	Tidal volume metrics	3.0
5	string	5% of tidal volume in liters	4.1
95	string	95% of tidal volume in liters	3.0
50	string	Median tidal volume in liters	3.0
max	string	Maximum tidal volume in liters	3.0
SPO2()	object	SpO ₂ metrics	3.3
min	string	minimum	3.6
5	string	5%	4.1
95	string	95%	3.3
50	string	50%	3.3
minutesBelow88percent	string	SpO ₂ minutes under the 88% threshold	3.3
secondsBelowDynamic Threshold	string	SpO ₂ seconds under dynamic threshold	3.6
alveolarVentilation()	object	Alveolar ventilation metrics	3.6
5	string	5% of alveolar ventilation in liters per minute	4.1
95	string	95% of alveolar ventilation in liters per minute	3.6
max	string	Maximum alveolar ventilation in liters per minute	3.6
50	string	Median alveolar ventilation in liters per minute	3.6
spontTriggBreaths	string	Percentage of spontaneous triggered breaths	3.0
spontCycledBreaths	string	Percentage of spontaneous cycled breaths	3.0
respEvents ()	object	Respiratory event indices recorded over the session	2.2
AHI	string	Apnea–hypopnea index in events per hour	2.2
HI	string	Hypopnea index in events per hour	2.2
AI	string	Apnea index in events per hour	2.2
OAI	string	Obstructive apnea index in events per hour	2.2
CAI	string	Central apnea index in events per hour	2.2
UAI	string	Unknown apnea index in events per hour	2.2
ODI	string	Oxygen Desaturation Index	3.6
CSR	string	Cheyne Stokes Respiration in minutes	3.0

Field Name (JSON Name)	Type/Format	Description	From Rev:
RERA	string	Respiratory Effort Related Arousal in events per hour	3.0
patientInterface ()	object	Climate control items for the session. Note: Only returned for Air™ 10 devices.	3.0
humidifier	string	Type of humidifier used on this day – Endcap, internal, or external	3.0
heatedTube	string	Type of heated tube used on this day – None, 15mm (ClimateLine) or 19mm (ClimateLine Max)	3.0
ambHumidity	string	Absolute ambient humidity for the session in milligrams of water per liter. Note: Only returned if humidification was used for the session.	3.0
humLevel	string	The humidity level of the humidifier. Note: Only available for High Flow Therapy (HFT) devices.	28.0
temp	string	The temperature of air delivered by the heated air tube in degrees Celsius. Note: Only available for High Flow Therapy (HFT) devices.	28.0

Example 1 (S9 AutoSet in CPAP mode):

```
{
  "Response": {
    "responseCode": "1004000",
    "responseDesc": "SUCCESS",
    "patientData": {
      "ecn": "6fbbb12b-093e-4ab3-af9a-1071a13c4c02",
      "totalUsageHoursOnDevice": "789",
      "sessionData": [
        {
          "receiptTime": "2014-03-02T00:00:00.000",
          "sessionDate": "2014-02-28",
          "deviceID": {
            "device": {
              "serialNo": "87021125448",
              "deviceType": "026001",
              "deviceTypeDesc": "S9 AutoSet"
            },
            "mode": "CPAP"
          },
          "usage": {
            "duration": "661",
            "maskOn": [
              "2014-02-28_20:25:00",
              "2014-02-28_22:58:00",
              "2014-03-01_04:02:00",
              "2014-03-01_09:37:00"
            ],
            "maskOff": [
              "2014-02-28_21:36:00",
              "2014-03-01_03:39:00",
              "2014-03-01_06:48:00",
              "2014-03-01_12:00:00"
            ]
          },
          "set": {
            "press": "18.6",
            "EPRTType": "OFF",
            "EPRLevel": "ONE"
          },
          "clinicalMetrics": {
            "leak": {
              "50": "0.283",
              "95": "0.308",
              "max": "0.321"
            }
          },
          "respEvents": {
            "AHI": "11.5",
            "AI": "15.2",
            "HI": "20.8",
            "CAI": "213.3",
            "OAI": "41.7",
            "UAI": "81.7"
          }
        },
        {
          "receiptTime": "2014-03-02T00:00:00.000",
          "sessionDate": "2014-03-01",
          "deviceID": {
            "device": {
              "serialNo": "87021125448",
              "deviceType": "026001",
              "deviceTypeDesc": "S9 AutoSet"
            },
            "mode": "CPAP"
          },
          "usage": {
            "duration": "144",
            "maskOn": [
              "2014-03-01_18:45:00"
            ]
          }
        }
      ]
    }
  }
}
```

```

        "maskOff": [
            "2014-03-01_21:09:00"
        ]
    },
    "set": {
        "press": "5.4",
        "EPRTYPE": "RAMP_ONLY",
        "EPRLevel": "THREE"
    },
    "clinicalMetrics": {
        "leak": {
            "50": "0.318",
            "95": "0.352",
            "max": "0.435"
        }
    },
    "respEvents": {
        "AHI": "44.7",
        "AI": "12.7",
        "HI": "6.7",
        "CAI": "178.7",
        "OAI": "143.5",
        "UAI": "67.7"
    }
}
    ]
}
}
}

```

Note: Data is fictional.

Example 2 (AirSense™ 10 AutoSet in CPAP Mode with SpO₂ data):

```
{
  "Response": {
    "responseCode": "1004000",
    "responseDesc": "SUCCESS",
    "patientData": {
      "ecn": "6fbbb12b-093e-4ab3-af9a-1071a13c4c02",
      "sessionData": [
        {
          "receiptTime": "2014-03-02T00:00:00.000",
          "sessionDate": "2014-02-28",
          "deviceID": {
            "device": {
              "serialNo": "98021125449",
              "deviceType": "036001",
              "deviceTypeDesc": "S10 AutoSet"
            },
            "mode": "CPAP"
          },
          "usage": {
            "duration": "661",
            "maskOn": [
              "2014-02-28_20:25:00",
              "2014-02-28_22:58:00",
              "2014-03-01_04:02:00",
              "2014-03-01_09:37:00"
            ],
            "maskOff": [
              "2014-02-28_21:36:00",
              "2014-03-01_03:39:00",
              "2014-03-01_06:48:00",
              "2014-03-01_12:00:00"
            ]
          },
          "set": {
            "press": "18.6",
            "EPRTType": "OFF",
            "EPRLLevel": "ONE"
          },
          "clinicalMetrics": {
            "leak": {
              "50": "0.283",
              "95": "0.308",
              "max": "0.321"
            },
            "SPO2": {
              "50": "95",
              "95": "98",
              "minutesBelow88percent": "59"
            }
          },
          "respEvents": {
            "AHI": "44.7",
            "AI": "12.7",
            "HI": "6.7",
            "CAI": "178.7",
            "OAI": "143.5",
            "UAI": "67.7"
          }
        }
      ]
    }
  }
}
```

Note: Data is fictional.

Errors:

Property	Value	
HTTP Response Type	Function of error – See table below	
Content Type	application/json	
Body	JSON “Response” Object – See table below	

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

Error	HTTP Response Type	responseCode
Requesting user does not have access to the patient	HTTP 401	1004001
The noOfDays is specified and endDate is missing	HTTP 400	1001002
There are no parameters defined and a session date is not specified		
Invalid parameter value	HTTP 400	1001003
Date is not in the correct format	HTTP 400	1001004
Badly formed data	HTTP 400	1001005

3.2 Get patient logged events

This API retrieves the five most recent days of patient Alarm, System and Settings Change events, listed with the most recent event first. **Note:** Alarm, System and Settings Change events are available to Astral and Stellar devices only. Alarm events are available to AirCurve 10 ST-A devices supporting alarms.

Property	Value
Method	GET
URLs	https://<host>/v1/patients/<ecn>/data/loggedEvents
	where:
	<i>ecn</i> is the patient unique identifier.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON “Response” Object – See table on the following page

The following fields are included in the JSON “Response” Object.

Field Name (JSON Name)	Type/Format	Description
responseCode	string	1004000
responseDesc	string	Unspecified - Debugging purposes only
patientLoggedEventsData()	object	Contains logged events
lastReceivedTime	string	Date and time logged events were received by server, in UTC time, format yyyy-mm-ddThh:mm:ssZ
LoggedEvents ([])	object array	Logged events for the patient
endDateTime	string	Date and Time event occurred in yyyy-mm-ddThh:mm:ss format
eventType	string	Type of event. May be one of: • System • Alarm • SettingsChange
details ()	object	Details of the event
systemCode	string	Used by System events. Code assigned to the system event. For available system codes and their descriptions, see "4 Appendix A. Supported system events" on page 23.
action	string	Used by Alarm events. May be one of: • Activate • Deactivate • Reset • Unknown
alarmType	string	The type of alarm. For available alarm types and their descriptions, see "5 Appendix B. Supported alarm types" on page 26.
alarmPriority	string	Used by Alarm events. May be one of: • Watchdog • High • Medium • Low • Unknown Not present if the alarmType is unknown. Note: Only available to Stellar and Astral devices.
alarmCode	string	Used by Alarm events. Code assigned to the alarm, used when alarmType is Unknown.
program	string	Used by Alarm and Settings Change events. The program the device was in when the alarm occurred. Note: Only available to Astral devices.
threshold()	object	Used by Alarm events. When available, used to provide threshold information regarding the alarm event.
thresholdCode	string	Used by Alarm events. The item that triggered the alarm event. Used when the thresholdType is not present.
thresholdType	string	Used by Alarm events. The item that triggered the alarm event.

Field Name (JSON Name)	Type/Format	Description
trigger()	object	Used by Alarm events. Provides information on the item triggering the alarm event.
value	string	Used by Alarm events. The value of the trigger. Note: This value should not be used if the units are unknown.
units	string	Used by Alarm events. The units of the triggered value.
settingType	string	Used by Settings Change events. The name of the setting that was changed.
settingCode	string	Used by Settings Change events. Code assigned to the settings, used when settingType is Unknown.
change()	object	Used by Settings Change events. Details of the settings change.
value	string	Used by Settings Change events. The value of the setting after the change. Note: This value should not be used if the units are unknown.
previous	string	Used by Settings Change events. The value of the setting before the change. Note: This value should not be used if the units are unknown.
units	string	Used by Settings Change events. If known, the units of the setting.

Example:

```
{
  "Response": {
    "responseCode": "1006000",
    "responseDesc": "SUCCESS",
    "deviceLoggedEventsData": {
      "loggedEvents": [
        {
          "endDateTime": "2016-04-14T20:47:44",
          "eventType": "SettingsChange",
          "details": {
            "program": "1",
            "settingType": "Val.TiMin",
            "change": {
              "value": "1.8",
              "previous": "2.2",
              "units": "s"
            }
          }
        },
        {
          "endDateTime": "2016-04-14T20:47:03",
          "eventType": "System",
          "details": {
            "systemCode": "E-020"
          }
        },
        {
          "endDateTime": "2016-04-14T19:10:15",
          "eventType": "Alarm",
          "details": {
            "action": "Activate",
            "alarmType": "BLOCKED_TUBE_ALARM",
            "alarmPriority": "High",
            "program": "1",
            "threshold": {
              "thresholdCode": "S-000",
              "trigger": {
                "value": "700",
                "units": "Unknown"
              }
            }
          }
        }
      ]
    }
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Type	Function of error – See table below
Content Type	application/json
Body	JSON “Response” Object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

Error	HTTP Response Type	responseCode
Requesting user does not have access to the patient	HTTP 401	1002001

3.3 Subscription/Notification interfaces

The following interface methods allow an integrator to subscribe to a patient. Following a subscription, you receive notifications when data arrives for that patient. This API call should only be implemented after the Notification Endpoint has been configured by the ResMed Operations Team and the Integration Team has authorized its usage. Note that this API supports S9, Air Solutions, Stellar and Astral devices.

Subscribing to a patient record for new summary data

Property	Value
Method	POST
URL	https://<host>/v1/subscriptions/patients/{ecr}/data/sessions
POST content	<Empty>

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON Object – See table below

Note: This response does not include the response code and response description. This information is being phased out from success calls as it is redundant.

Field Name (JSON Name)	Type/Format	Description
UUID	string	Unique identifier for the subscription
resource	string	The resource that was subscribed to

Example:

```
{
  "resource": "patients/1272a067-ca5e-4ba6-86ce-01ff75543d9a/data/sessions",
  "UUID": "17ae91dc-75e2-4d28-b90a-53410543ca76"
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" Object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

Error Description	Response Type	responseCode
Requesting user does not have access to the patient	HTTP 401	1013001

3.3.1 Patient notifications (AVX to IA)

This message is pushed to the IA on the following condition:

- Data arrives for a patient, and the IA is subscribed to that patient as per "Subscribing to a patient record for new summary data" on page 19.

Also note:

- This is a simple notifications interface that sends a notification for every day of data that is received wirelessly, even if multiple days are sent on the same day for the same patient. This is because wireless devices send each summary day individually to AirView.
- The integrator can choose to consolidate the notifications and reduce the number of calls to AVX.
- An exception to this is card download, for which only one notification is sent.

Property	Value
Method	POST
URL	As provided by the integrating application upon registration. See the section called <i>Adding an integrator to AirView/AVX (including authentication tokens)</i> in the <i>AVX API Technical Specification for Patient Management</i> .
POST content	A JSON object – See table below

A JSON object containing the following element(s) forms the body of the POST request.

Field Name (JSON Name)	Type	Description
resource	string	The resource that was modified – See table on the following page
UUID	string	Unique identifier for the subscription. The UUID is required to delete a subscription.
updatedAtOn	string	Time that resource was modified in System Time with format: YYYY-MM-DD_HH:MM:SS

The following table contains values for resources:

Resource subscribed to	Resource in notification	Description
patients/{ecn}/data/sessions	patients/{ecn}/data/sessions	Indicates that data was received for patient with ecn. Note: It is anticipated that an IA can use this ecn to request data for this patient receivedOn the updatedOn data provided above. An indication of which sessionDates were received for the patient is not currently supported.

Example:

```
{
  "resource": "patients/6fbbb12b-093e-4ab3-af9a-1071a13c4c02/data/sessions",
  "UUID": "17ae91dc-75e2-4d28-b90a-53410543ca76",
  "updatedOn": "2014-06-05 10:25:15"
}
```

Note: Data is fictional.

3.3.2 Integrator responses with corresponding AVX behavior

The following table captures the possible responses from the integrator, and the corresponding AVX behavior.

Response	Description of AVX behavior
HTTP 200	Treated as success. AVX progresses to the next message in the queue for that integrator if one exists.
HTTP 400	AVX does not attempt to resend this message, and progresses to the next message in the queue for that integrator if one exists.
Any other response, or no response	AVX resends the message at 5-second intervals until either an HTTP 200 or an HTTP 400 is received. After three days, the notification message is transferred to a DLQ for two weeks before removal.

Notifications are sent serially for each integrator. Subsequent notifications are not processed until the current notification is responded to with either an HTTP 200 or HTTP 400 response.

It is expected that troubleshooting will be required if HTTP 400 responses are being received. If an integrator sends an HTTP 400, then it is advisable to include an error description in the body of the message to aid with such troubleshooting. Through development and test, HTTP 400 errors are expected to be minimal.

ResMed tries to send the notification every five seconds until an HTTP status code of 200 (success) or 400 (failure) is received.

If any other response is received, or a response is not received, ResMed retries the notification.

If the integrator system does not respond or return an expected result, ResMed continues to try to send the notification for a period of three days. After this time has elapsed, ResMed moves the notification messages to a dead letter queue (DLQ). Notifications remain on this DLQ for a period of two weeks, then they are removed. **Note:** Notifications cannot be retrieved after they are removed from the DLQ.

Notifications can be retrieved from the DLQ, however this is not an automatic process. The integrator must contact ResMed to have these messages re-sent to their system. ResMed will respond within a reasonable period after the request is made.

Note: Since notifications can and will be sent 24-hours a day, integrators should inform ResMed in advance when a downtime is anticipated so that the sending of notifications can be turned off until the system is back online.

4 Appendix A. Supported system events

4.1 Astral system events

System Code	Description
E-001	Passed Learn Circuit
E-007	SpO ₂ connected
E-008	SpO ₂ disconnected
E-009	O ₂ sensor connected
E-010	O ₂ sensor disconnected
E-018	Ventilation started
E-019	Ventilation stopped
E-025	Ventilator powered on
E-026	Ventilator powered off
E-048	Skipped Learn Circuit
E-049	Failed Learn Circuit
E-050	Using internal battery
E-051	Using external power source
E-052	USB stick inserted
E-053	USB stick removed
E-056	PEEP turbine service due
E-057	Main blower service due
E-058	Transfer data
E-059	Data transfer complete
E-060	Data transfer interrupted
E-061	Unable to transfer data
E-062	PEEP turbine nearing service
E-063	Main blower nearing service
E-064	35,000 hours service due
E-067	Patient data logging disabled
E-068	Enter clinical mode
E-069	Exit clinical mode
E-070	Patient data logging enabled
E-072	Passed Learn Circuit with warning
E-073	Device auto powered off
E-074	Logged data corrupt
E-075	Using external power source
E-076	Using external power source
E-077	Using external power source
E-078	Using external power source
E-079	Settings copy 1 corrupt
E-080	Settings copy 2 corrupt
E-081	Device self-test failed. Circuit not tested.
E-082	Canceled Learn Circuit

4.2 Stellar system events

System Code	Description
E-001	Learn Circuit passed
E-002	Learn Circuit canceled
E-003	iVAPS Learn Mode passed
E-004	iVAPS Learn Mode canceled
E-005	USB connected
E-006	USB disconnected
E-007	XPOD connected
E-008	XPOD disconnected
E-009	O ₂ sensor connected
E-010	O ₂ sensor disconnected
E-011	Reminder set
E-013	Humidifier connected
E-014	Humidifier disconnected
E-015	Internal battery active
E-016	Internal battery inactive
E-018	Mask on
E-019	Mask off
E-020	Erase data
E-021	Reset to factory default
E-022	Internal battery reminder set
E-023	Internal battery reminder cleared
E-024	Incorrect shutdown
E-025	FG turned on
E-026	FG turned off
E-027	System startup failure: FFS
E-028	System startup failure: PRESSURETABLE
E-029	System startup failure: FLOWTABLE
E-030	System startup failure: HOSEDROPTABLE
E-031	System startup failure: GUI
E-032	System startup failure: CONFIGURATIONSETTINGS
E-033	System startup failure: PATIENTSETTINGS
E-034	System startup failure: MOTORTTEST
E-036	System startup failure: PCBINFO
E-037	System startup failure: ALLOCPATIENTINFO
E-038	System startup failure: STARTTASK
E-039	System startup failure: LoggingNotAvailable
E-040	System startup failure: LoggingFilesCorrected
E-041	System detected event not logged
E-042	FG emergency shutdown
E-043	FG auto restart
E-044	System startup failure: BASICLOGGING

System Code	Description
E-045	System startup failure: READWRITEFILESYSYSTEM
E-046	FG 10s powerbutton shutdown
E-047	AC connected
E-048	Ramp down started
E-083	Alarm muted
E-084	Alarm mute cancelled
E-085	High Resistance Circuit
E-086	Started Learn Circuit
E-087	Unable to learn circuit. (Error 204)
E-088	Unable to learn circuit. (Error 205)
E-090	Unable to learn circuit. (Error 420)
E-091	Unable to learn circuit. (Error 504)
E-092	Unable to learn circuit. (Error 505)
E-093	Unable to learn circuit. (Error 512)
E-095	Self Test Failed. (Error 001)
E-096	Self Test Failed. (Error 104)
E-097	Self Test Failed. (Error 105)
E-098	Self Test Failed. (Error 106)
E-099	Self Test Failed. (Error 113)
E-100	Self Test Failed. (Error 121)
E-101	Self Test Failed. (Error 122)
E-102	Self Test Failed. (Error 123)
E-103	Self Test Failed. (Error 124)
E-104	Self Test Failed. (Error 125)
E-105	Self Test Failed. (Error 206)
E-106	Self Test Failed. (Error 404)
E-107	Self Test Failed. (Error 405)
E-108	Self Test Failed. (Error 406)
E-109	Self Test Failed. (Error 409)
E-110	Self Test Failed. (Error 415)
E-111	Self Test Failed. (Error 426)
E-112	Self Test Failed. (Error 506)
E-113	Oxygen sensor not detected.
E-114	Oxygen sensor failure. (Error 303)
E-115	Unable to calibrate expiratory flow sensor. (Error 600)

5 Appendix B. Supported alarm types

5.1 Astral alarms

Alarm Type	Description
INTERNAL_BATTERY_DEPLETED_ALARM	Critically low battery alarm
TOTAL_POWER_FAILURE_ALARM	Total power failure alarm
LOW_PEAK_INSPIRATORY_PRESSURE_ALARM	Low pressure alarm
LOW_EXPIRATORY_TIDAL_VOLUME_ALARM	Low expiratory tidal volume alarm
LOW_INSPIRATORY_TIDAL_VOLUME_ALARM	Low inspiratory tidal volume alarm
HIGH_INSPIRATORY_TIDAL_VOLUME_ALARM	High inspiratory tidal volume alarm
LOW_EXPIRATORY_MINUTE_VOLUME_ALARM	Low expiratory minute volume alarm
HIGH_EXPIRATORY_MINUTE_VOLUME_ALARM	High expiratory minute volume alarm
LOW_INSPIRATORY_MINUTE_VOLUME_ALARM	Low inspiratory minute volume alarm
HIGH_INSPIRATORY_MINUTE_VOLUME_ALARM	High inspiratory minute volume alarm
INTERNAL_BATTERY_LOW_ALARM	Low internal battery alarm
LOW_PEEP_ALARM	Low PEEP alarm
HIGH_PEEP_ALARM	High PEEP alarm
LOW_RESPIRATORY_RATE	Low respiratory rate alarm
EXTERNAL_DC_POWER_DISCONNECT_ALARM	Using internal battery alarm
POWERED_ON_BATTERY_ALARM	Using internal battery alarm
INTERNAL_BATTERY_INOPERABLE_ALARM	Battery inoperable alarm
HIGH_RESPIRATORY_RATE	High respiratory rate alarm
TEST_ALARM_1	TEST_ALARM_1 alarm
TEST_ALARM_2	TEST_ALARM_2 alarm
TEST_ALARM_3	TEST_ALARM_3 alarm
TEST_ALARM_4	TEST_ALARM_4 alarm
TEST_ALARM_5	TEST_ALARM_5 alarm
TEST_ALARM_6	TEST_ALARM_6 alarm
PROGRAM_DATA_CORRUPT_ALARM	System fault - 65 alarm
SOFTWARE_TASK_FAILURE_ALARM	System fault - 74 alarm
PB_NOT_CALIBRATED_ALARM	System fault - 75 alarm
ALARM_SYSTEM_FAILURE_INVALID_HAC_RESPONSE_ALARM	System fault - 82 alarm
HIGH_EXPIRATORY_TIDAL_VOLUME_ALARM	High expiratory tidal volume alarm
LOW_FIO2_ALARM	Low FiO ₂ alarm
HIGH_FIO2_ALARM	High FiO ₂ alarm
HIGH_INSPIRATORY_PRESSURE_ALARM	High pressure alarm
SUSTAINED_HIGH_INSPIRATORY_PRESSURE_ALARM	High pressure alarm
DATA_TRANSFER_CRC_FAILURE_ALARM	System fault - 98 alarm
OUTLET_PRESSURE_MEASUREMENT_INCORRECT_ALARM	System fault - 101 alarm
MONITORING_FAILURE_PRESSURE_SENSOR_1_ALARM	System fault - 102 alarm
MONITORING_FAILURE_PRESSURE_SENSOR_2_ALARM	System fault - 103 alarm

Alarm Type	Description
HIGH_EXPIRATORY_PRESSURE_ALARM	Obstruction / High pressure alarm
APNEA_ALARM	Apnea alarm
TRIGGER_PRESSURE_MEASUREMENT_INCORRECT_IN_USE_ALARM	System fault - 108 alarm
MONITORING_FAILURE_CHECK_CIRCUIT_ALARM	Circuit fault alarm
LOW_SPO2_ALARM	Low SpO ₂ alarm
HIGH_SPO2_ALARM	High SpO ₂ alarm
MONITORING_INVALID_ATMOSPHERIC_PRESSURE_ALARM	System fault - 115 alarm
MONITORING_INVALID_AMBIENT_TEMPERATURE_ALARM	System fault - 116 alarm
INVALID_REQUEST_TO_STOP_THERAPY_ALARM	System fault - 117 alarm
NON_VENTED_MASK_ALARM	Non-vented mask alarm
PROXIMAL_PRESSURE_LINE_DISCONNECT_ALARM	Pressure line disconnected alarm
BAD_CIRCUIT_CONFIGURATION_ALARM	Incorrect circuit attached alarm
LOW_PULSE_RATE_ALARM	Low pulse rate alarm
HIGH_PULSE_RATE_ALARM	High pulse rate alarm
POWERED_ON_DEPLETED_BATTERY_ALARM	Critically low battery alarm
FAULT_DETECTED_ALARM	Last self-test failed alarm
PROXIMAL_PRESSURE_MEASUREMENT_INCORRECT	System fault - 125 alarm
NO_OXIMETRY_MONITORING_ALARM	No SpO ₂ monitoring alarm
NO_O2_MONITORING_ALARM	No FiO ₂ monitoring alarm
VENTED_EXPIRATORY_MODULE_MISSING_ALARM	Ventilation not started, incorrect adapter alarm
BLOWER_TEMPERATURE_MONITORING_FAILURE_ALARM	System fault - 131 alarm
TRIGGER_PRESSURE_MEASUREMENT_INCORRECT_NOT_IN_USE_ALARM	System fault - 132 alarm
PEEP_VALVE_PRESSURE_MEASUREMENT_INCORRECT_ALARM	System fault - 133 alarm
NVM_CORRUPTION_NON_CRITICAL_DATA_ALARM	System fault - 134 alarm
MAIN_BLOWER_SOFTWARE_OVERTEMPERATURE_ALARM	Device overheating alarm
HIGH_LEAK_ALARM	High leak alarm
LOSS_OF_COMMUNICATION_WITH_THE_PB_ALARM	System fault - 137 alarm
PB_REFUSE_BREATH_STATE_CHANGE_ALARM	System fault - 138 alarm
LOSS_OF_CONTROL_DATA_ALARM	System fault - 139 alarm
ALARM_SYSTEM_FAILURE_WRONG_PRIORITY_ALARM	System fault - 140 alarm
INVALID_REQUEST_TO_ENTER_HAM_ALARM	System fault - 141 alarm
INVALID_REQUEST_TO_PERFORM_CIRCUIT_TEST_ALARM	System fault - 142 alarm
FAILURE_TO_START_VENTILATION_ALARM	System fault - 143 alarm
MAXIMUM_INSPIRATORY_OR_EXPIRATORY_TIME_LIMIT_EXCEEDED_ALARM	System fault - 144 alarm
MB_SW_WATCHDOG_ALARM	System fault - 145 alarm
MAIN_BLOWER_FAILURE_STALLED_ALARM	System fault - 147 alarm
MAIN_BLOWER_FAILURE_HARDWARE_OVER_TEMPERATURE_ALARM	System fault - 148 alarm
MAIN_BLOWER_FAILURE_OVER_CURRENT_ALARM	System fault - 149 alarm
ELECTRO_VALVE_FAILURE_OVER_CURRENT_ALARM	System fault - 150 alarm
EXPIRATORY_FLOW_SENSOR_NOT_CALIBRATED_ALARM	Flow sensor not calibrated alarm

Alarm Type	Description
ABNORMAL_RESTART_ALARM	Safety reset complete alarm
VENTILATION_STOP_ALARM	Ventilation stopped alarm
COOLING_FAN_FAILURE_ALARM	Cooling fan fault alarm
SUPER_CAPACITOR_FAULT_ALARM	System fault - 179 alarm
BATTERY_CHARGER_FAULT_ALARM	System fault - 180 alarm
BATTERY_COMMUNICATION_LOST_INTERNALLY_POWERED_ALARM	Battery comms lost alarm
MAIN_PCB_OVERTEMPERATURE_ALARM	Device overheating alarm
OUTLET_SENSOR_DATA_IMPLAUSIBLE_ALARM	System fault - 185 alarm
LOSS_OF_COMMUNICATION_WITH_EXPIRATORY_FLOW_SENSOR	Flow sensor fault alarm
SAFETY_ASSERT_SYSTEM_FAULT_ALARM	System fault - 188 alarm
LOSS_OF_COMMUNICATION_WITH_THE_MB_ALARM	System fault - 189 alarm
CRITICAL_OUTLET_PRESSURE_SENSOR_FAILURE_ALARM	System fault - 190 alarm
LOSS_OF_COMMUNICATION_WITH_OUTLET_FLOW_SENSOR_ALARM	System fault - 191 alarm
POWER_TO_SENSOR_BOARD_COMPROMISED_ALARM	System fault - 192 alarm
PERSISTENT_SOFTWARE_FAULT_ALARM	System fault - 195 alarm
INCORRECT_SETTINGS_ALARM	System fault - 196 alarm
OUTLET_SENSOR_STUCK_ALARM	System fault - 197 alarm
MISSING_PREVENTILATION_DATA_ALARM	System fault - 198 alarm
CALIBRATION_CORRUPTED_ALARM	System fault - 199 alarm
BLOWER_TEMPERATURE_SENSOR_DISCONNECT_ALARM	System fault - 200 alarm
OVERPRESSURE_PROTECTION_ALARM	High pressure protection alarm
INTERNAL_BATTERY_INOPERABLE_EMPTY_ALARM	Critically low battery alarm
MB_WATCHDOG_ALARM	System fault - 218 alarm
PB_WATCHDOG_ALARM	System fault - 219 alarm
MB_AND_PB_WATCHDOG_ALARM	System fault - 220 alarm
PEEP_BLOWER_FAILURE_ALARM	PEEP Blower Failure alarm
BATTERY_PACKET_ERROR_CHECKING_FAULT_ALARM	System fault - 225 alarm
DEVICE_OVERHEATING_ALARM	Device overheating alarm
EXTERNAL_BATTERY_1_INOPERABLE	Battery 1 fault alarm
EXTERNAL_BATTERY_2_INOPERABLE	Battery 2 fault alarm
EXTERNAL_BATTERY_1_COMMUNICATION_LOST	Battery 1 fault alarm
EXTERNAL_BATTERY_2_COMMUNICATION_LOST	Battery 2 fault alarm
DISCONNECTION_ALARM	Circuit disconnection alarm
INTERNAL_BATTERY_DEGRADED_ALARM	Internal Battery Degraded alarm
OBSTRUCTION_ALARM	Obstruction alarm
HIGH_PRESSURE_ALARM	High pressure alarm
SUSTAINED_HIGH_PRESSURE_ALARM	High pressure alarm

5.2 Stellar alarms

Alarm Type	Description
OVERPRESSURE_ALARM	Over pressure alarm
INTERNAL_BATTERY_EMPTY_ALARM	Internal battery empty alarm
HIGH_PRESSURE_ALARM	High pressure alarm
NON_VENTED_MASK_ALARM	Non-vented mask alarm
BLOCKED_TUBE_ALARM	Blocked tube alarm
SYSTEM_FAILURE_[6]_ALARM	System failure [6] alarm
SYSTEM_FAILURE_[7]_ALARM	System failure [7] alarm
SYSTEM_FAILURE_[8]_ALARM	System failure [8] alarm
SYSTEM_FAILURE_[9]_ALARM	System failure [9] alarm
HIGH_TEMPERATURE_[10]_ALARM	High temperature [10] alarm
HIGH_TEMPERATURE_[11]_ALARM	High temperature [11] alarm
HIGH_TEMPERATURE_[12]_ALARM	High temperature [12] alarm
LOW_PRESSURE_ALARM	Low pressure alarm
OPEN_CIRCUIT_ALARM	Circuit disconnected alarm
LOW_MINUTE_VENTILATION_ALARM	Low minute ventilation alarm
APNEA_ALARM	Apnea alarm
HIGH_BREATH_RATE_ALARM	High respiratory rate alarm
LOW_BREATH_RATE_ALARM	Low respiratory rate alarm
INTERNAL_BATTERY_LOW_ALARM	Low internal battery alarm
SYSTEM_FAILURE_[21]_ALARM	System failure [21] alarm
SYSTEM_FAILURE_[22]_ALARM	System failure [22] alarm
HIGH_TEMPERATURE_[23]_ALARM	High temperature [23] alarm
HIGH_LEAK_ALARM	High leak alarm
SYSTEM_FAILURE_[25]_ALARM	System failure [25] alarm
EXTERNAL_BATTERY_USE_ALARM	External battery use alarm
INTERNAL_BATTERY_USE_ALARM	Internal battery use alarm
KEYPAD_FAILURE_ALARM	Keypad failure alarm
FINGER_SENSOR_FAILURE_ALARM	Finger sensor failure alarm
XPOD_DISCONNECTED_ALARM	XPOD disconnected alarm
HIGH_FIO2_ALARM	High FiO ₂ alarm
LOW_FIO2_ALARM	Low FiO ₂ alarm
FIO2_CELL_INTEGRITY_ALARM	O ₂ sensor disconnected alarm
SYSTEM_FAILURE_[38]_ALARM	System failure [38] alarm
LOW_SPO2_ALARM	Low SpO ₂ alarm
MOTOR_TEMP_ATTENTION_ALARM	Temperature attention [42] alarm
INTERNAL_BATT_TEMP_ATTENTION_ALARM	Temperature attention [43] alarm
POWER_SUPPLY_TEMP_ATTENTION_ALARM	Temperature attention [44] alarm
PCB_TEMP_ATTENTION_ALARM	Temperature attention [45] alarm

5.3 AirCurve 10 ST-A alarms

Alarm Type	Description
NoAlarm	No alarm
MotorFault	Motor fault
HighPressureFault	High pressure fault
HighTemperatureFault	High temperature fault
SensorFault	Sensor fault
SystemError	System error
CalToDefault	Calibration reset to default
SettingsToDefault	Settings reset to default
OverVoltage	Over voltage fault
AlarmUnitAlarm	Alarm unit fault
HumidifierTubEndcapOpen	Humidifier tub/endcap open
TubeBlocked	Tube blocked
TubeOpen	Tube disconnected
HighLeak	High leak
NonVentedMask	Non-vented mask
LowMinuteVentilation	Low min ventilation
ApneaAlarm	Apnoea alarm
SpO2Alarm	Low SpO ₂
SpO2Disconnection	SpO ₂ disconnected
SpO2BadData	Invalid SpO ₂ data

6 Appendix C. Supported devices

6.1 S9 devices

The following table includes the valid device types required to add an S9 device to a patient file.

S9 device name	deviceType
S9 AutoSet	026001
S9 Elite	026002
S9 Escape	026003
S9 Escape Auto	026004
S9 VPAP Auto	026009
VPAP S (S9)	026011
VPAP ST (S9)	026012
VPAP Adapt (S9)	026013
VPAP Adapt (S9 w/ ASVAuto)	026016
VPAP ST-A	026018
VPAP COPD	026023

6.2 All other devices

For devices manufactured after the S9, only the serial number and device number are used to add the device to a patient file.

For a list of all supported devices, see the Compatible devices topic in the AirView Help.



ResMed Corp

9001 Spectrum Center Boulevard San Diego CA 92123 USA

See www.resmed.com for other ResMed locations worldwide.

For patent and other intellectual property information, see www.resmed.com/ip.

AirCurve, AirSense, AirView, Astral, S9 and Stellar are trademarks and/or registered trademarks of the ResMed family of companies. 2281358/8 2022-10

ResMed.com